# DIFFERENTIAL COHOMOLOGY SEMINAR 8

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The aim of this talk is to review twisted cohomology theory with the aim of later discussing twisted differential cohomology theories [BG16]. For these talks the main source is [ABG+14, ABG18].

## 1. Twisted Cohomology

Let $R$ be a ring spectrum, meaning a monoid object in the $\infty$-category of spectra $\mathcal{S}\mathrm{p}$. From this we get a presentable stable $\infty$-category $\mathcal{M}\mathrm{od}_R$ of left $R$-module spectra. Objects therein are morphisms of the form $R \wedge M \to M$ satisfying the usual associativity and unit conditions up to coherent homotopies.

*Remark* 1.1. Every stable category $\mathcal{C}$ is enriched over spectra. Given two objects $C, D$, we will denote by $\mathcal{H}\mathrm{om}_{\mathcal{C}}(C, D)$ the hom-spectrum and by $\mathrm{Hom}_{\mathcal{C}}(C, D) = \Omega^\infty \mathcal{H}\mathrm{om}_{\mathcal{C}}(C, D)$ the underlying hom-space.

Note we have an adjunction

$$\mathcal{S}\mathrm{p} \underset{\mathcal{H}\mathrm{om}_{\mathcal{M}\mathrm{od}_R}(R,-)}{\overset{R\wedge -}{\rightleftarrows}} \mathcal{M}\mathrm{od}_R \ ,$$

where the right adjoint is equivalent to the forgetful functor.

**Definition 1.2.** Let $R$ be a ring spectrum. An *$R$-line* is an $R$-module $L$ such that $L \simeq R$.

**Definition 1.3.** Let $\mathcal{L}\mathrm{ine}_R$ be the full sub-$\infty$-groupoid of $\mathcal{M}\mathrm{od}_R$ spanned by the $R$-lines.

By construction, $\mathcal{L}\mathrm{ine}_R$ is equivalent to the category with a single object $R$ and hom-space $GL_1R$, the $\infty$-group of $R$-linear automorphisms of $R$. Notice that $GL_1(R) \subseteq \mathrm{Hom}_{\mathcal{M}\mathrm{od}_R}(R, R) \simeq \mathrm{Hom}_{\mathcal{S}\mathrm{p}}(\mathbb{S}, R) = \Omega^\infty R$.

**Lemma 1.4.** *$GL_1(R)$ fits into the pullback square*

$$\begin{tikzcd} GL_1(R) \arrow[r] \arrow[d] & \Omega^\infty R \arrow[d] \\ \pi_0(R)^\times \arrow[r] & \pi_0(R) \end{tikzcd}$$

*In particular, the inclusion $GL_1(R) \to \Omega^\infty R$ induces an isomorphism on $n$-homotopy groups, for all $n \geq 1$.*

*Proof.* $\pi_0(R) \simeq \mathrm{Hom}_{\mathrm{Ho}\mathcal{M}\mathrm{od}_R}(R, R)$, where $\mathrm{Ho}\mathcal{M}\mathrm{od}_R$ is the homotopy category of $R$-modules, the right-vertical arrow corresponds to $\mathrm{Hom}_{\mathcal{M}\mathrm{od}_R}(R, R) \to \mathrm{Hom}_{\mathrm{Ho}\mathcal{M}\mathrm{od}_R}(R, R)$ mapping a morphism to its homotopy class, and $\pi_0(R)^\times \subseteq \mathrm{Hom}_{\mathrm{Ho}\mathcal{M}\mathrm{od}_R}(R, R)$ is the set of isomorphisms. Finally, a morphism in a $\infty$-category $\mathcal{C}$ is an equivalence if and only if its homotopy class is an isomorphism in $\mathrm{Ho}\mathcal{C}$. $\square$

**Definition 1.5.** Let $X$ be a space. Denote by $\mathcal{M}\mathrm{od}_R(X)$ the $\infty$-category of $R$-module spectra parametrized over $X$, i.e. the functor category $\mathcal{F}\mathrm{un}(X^{\mathrm{op}}, \mathcal{M}\mathrm{od}_R)$.

**Definition 1.6.** Let $X$ be a space. Denote by $\mathcal{L}\mathrm{ine}_R(X)$ the $\infty$-groupoid of $R$-line spectra parametrized over $X$, i.e. the functor category $\mathcal{F}\mathrm{un}(X^{\mathrm{op}}, \mathcal{L}\mathrm{ine}_R)$.

Since $\mathcal{L}\mathrm{ine}_R \simeq BGL_1(R)$, functors $X^{\mathrm{op}} \to \mathcal{L}\mathrm{ine}_R$ classify $GL_1(R)$-principal bundles over $X$. In this sense, elements of $\mathcal{L}\mathrm{ine}_R(X)$ are $\infty$-local systems.

**Example 1.7.** Let $R_X \colon X^{\mathrm{op}} \to * \to \mathcal{L}\mathrm{ine}_R$ be the constant functor with value $R$.

We now proceed to the Thom construction. We introduce it to full generality.

*Date*: 12.11.2025 & 26.11.2025.

**Definition 1.8** (Thom spectrum)**.** The *Thom $R$-module spectrum* is the functor

$$R\colon \mathcal{S}/\mathcal{M}\mathrm{od}_R \to \mathcal{M}\mathrm{od}_R,$$

which sends $f\colon X^{\mathrm{op}} \to \mathcal{M}\mathrm{od}_R$ to the $R$-module spectrum $R^f$ given by colimit of $f$.

*Remark* 1.9. The colimit of a functor $F : \mathcal{C} \to \mathcal{X}$ is the left Kan extension of $F$ along the terminal functor $p : \mathcal{C} \to *$. If $\mathcal{X}$ is cocomplete, a functor $\pi : \mathcal{C} \to \mathcal{D}$ induces a pullback functor $\pi^* : \mathcal{F}\mathrm{un}(\mathcal{D}, \mathcal{X}) \to \mathcal{F}\mathrm{un}(\mathcal{C}, \mathcal{X})$, which is cocontinuous, therefore using left Kan extension along $\pi$ we can construct a left adjoint $\pi_!$.

Let us have a look at some examples of Thom spectra.

**Example 1.10** (Trivial twist)**.** We begin by identifying the Thom spectrum of $R_X$. By the universal property of colimits, the Thom spectrum functor is left adjoint to the functor $\mathcal{M}\mathrm{od}_R \to \mathcal{S}/\mathcal{M}\mathrm{od}_R$ sending a module $E$ to the corresponding functor $* \to \mathcal{M}\mathrm{od}_R$. In particular, the Thom spectrum functor is cocontinuous. Next, consider a $R$-module $E$ and the functor $\mathcal{S} \to \mathcal{S}/\mathcal{M}\mathrm{od}_R$ sending a space $X$ to the constant functor $E_X : X^{\mathrm{op}} \to \mathcal{M}\mathrm{od}_R$ with value $E$. This functor is also cocontinuous, by the way colimits are calculated in $\mathcal{S}/\mathcal{M}\mathrm{od}_R$. Therefore, the composition

$$\mathcal{S} \longrightarrow \mathcal{S}/\mathcal{M}\mathrm{od}_R \longrightarrow \mathcal{M}\mathrm{od}_R \tag{1.11}$$

is cocontinuous, and so completely determined by the value of the one-point space $*$, which is $E$. On the other hand, the following composition

$$\mathcal{S} \xrightarrow{\Sigma^\infty_+} \mathcal{S}\mathrm{p} \xrightarrow{E\wedge -} \mathcal{M}\mathrm{od}_R \tag{1.12}$$

is also cocontinuous and sends $*$ to $E$. We conclude that the colimit of $E_X$, for every space $X$, is equivalent to $E \wedge \Sigma^\infty_+ X$.

For the next example, we require some preliminary lemmas.

**Definition 1.13.** Given a vector bundle $V \to B$, the *Thom space* of $V$, denoted by $S^V$, is the homotopy cofiber of $V_0 \subseteq V$, where $V_0$ is the complement of the zero section.

**Example 1.14.** Let $B = *$ and $V = \mathbb{R}^n$, then $S^V$ is homotopy equivalent to the $n$-sphere $S^n$.

**Lemma 1.15.** *Let $\mathcal{C}$ be a (small) category and $y : \mathcal{C} \to \mathcal{F}\mathrm{un}(\mathcal{C}^{\mathrm{op}}, \mathcal{S})$ the $\infty$-categorical Yoneda embedding. The colimit of $y$ is the terminal pre-sheaf, i.e. the pre-sheaf with constant value $*$.*

*Proof.* Apply the density theorem to the terminal pre-sheaf. □

We now proceed to define twisted cohomology theories.

**Definition 1.16** (Twisted cohomology)**.** Let $R$ be a ring spectrum, $X$ be a space, $p\colon X \to *$ the terminal functor, and $f\colon X^{\mathrm{op}} \to \mathcal{L}\mathrm{ine}_R$ a $R$-line bundle over $X$. The *$f$-twisted $R$-cohomology of $X$* is defined as the mapping spectrum

$$R_f(X) \coloneqq \mathcal{H}\mathrm{om}_{\mathcal{M}\mathrm{od}_R}(Mf, R) \simeq \mathcal{H}\mathrm{om}_{\mathcal{M}\mathrm{od}_R(X)}(f, p^*R) \simeq \mathcal{H}\mathrm{om}_{\mathcal{M}\mathrm{od}_R(X)}(f, R_X).$$

Similarly, the *$f$-twisted $R$-homology of $X$* is defined as

$$R^f(X) \coloneqq \mathcal{H}\mathrm{om}_{\mathcal{M}\mathrm{od}_R}(R, Mf) \simeq Mf.$$

The $f$-twisted $R$-cohomology groups of $X$ are defined as the homotopy groups of $R_f(X)$, i.e.

$$R^n_f(X) \coloneqq \pi_0(\mathcal{H}\mathrm{om}_{\mathcal{M}\mathrm{od}_R}(Mf, \Sigma^n R)) \cong \pi_{-n}(\mathcal{H}\mathrm{om}_{\mathcal{M}\mathrm{od}_R(X)}(f, R_X)).$$

Similarly, the $f$-twisted $R$-homology groups of $X$ are defined as the homotopy groups of $R^f(X)$, i.e.

$$R^f_n(X) \coloneqq \pi_0(\mathcal{H}\mathrm{om}_{\mathcal{M}\mathrm{od}_R}(\Sigma^n R, Mf)) \cong \pi_n(Mf).$$

**Example 1.17** (Trivial twist)**.** If $f : X \to \mathcal{L}\mathrm{ine}_R$ factors through $*$, then $f$ factors as $X^{\mathrm{op}} \to * \to \mathcal{S}$, the constant factor with value $*$, and $R \wedge \Sigma^\infty_+(-) : \mathcal{S} \to \mathcal{M}\mathrm{od}_R$. The latter functor commutes with colimits, being a left adjoint, while the colimit of the latter is $X$ itself, then $Mf \simeq R \wedge \Sigma^\infty_+ X$. In particular, $f$-twisted $R$-cohomology and $R$-homology of $X$ reduce to ordinary (untwisted) $R$-cohomology and $R$-homology of $X$.

## 2. Examples of Twisted Cohomology

We now proceed to analyze several examples of twisted cohomology theories. This requires some preliminary lemmas.

**Lemma 2.1.** *Consider a space $X$ and the $\infty$-categorical Yoneda's embedding $y : X \to \mathcal{F}\mathrm{un}(X^{\mathrm{op}}, \mathcal{S})$. The colimit of $y$ is the terminal pre-sheaf on $X$, i.e. the pre-sheaf with constant value the one-point space.*

*Proof.* Let $S$ be a pre-sheaf on $X$, consider then the slice category $X_{/S}$ of pairs $(x, \phi)$, where $x$ is an object of $X$ and $\phi : y(x) \to S$. The density theorem for $\infty$-categories states that $S$ is equivalent to the colimit of $X_{/S} \to X \xrightarrow{y} \mathcal{F}\mathrm{un}(X^{\mathrm{op}}, \mathcal{S})$, the first map being the canonical projection. Take $S = *$, then $X_{/*} \to X$ is an equivalence, hence the claim. □

Let $G$ be a topological group and $BG$ the $\infty$-groupoid with a single object $*$ and hom-space $G$. The category $\mathcal{S}_G \coloneqq \mathcal{F}\mathrm{un}(BG, \mathcal{S})$ is equivalent to the category of $G$-spaces.

**Lemma 2.2.** *Consider $X = BG$, a $G$-space $f : X \to \mathcal{S}$ and its left Kan extension $f_! : \mathcal{S}_G \to \mathcal{S}$, then $f_! \simeq (- \times E)/G$, where $E = f(*)$.*

*Proof.* Evaluate at $*$, then $f_!(y(*)) = E$, by definition, and $y(*) \simeq G$, as $G$-spaces, hence $(y(*) \times E)/G \simeq (G \times E)/G \simeq E$. Since $f_!$ and $(- \times E)/G$ agree on representables and are colimit-preserving, they are equivalent. □

**Example 2.3.** Take the space $BO(n)$ and $f_n : BO(n) \to \mathcal{S}_*$ the $n$-sphere $S^n$ with $O(n)$-action coming from the one-point compactification of the regular action on $\mathbb{R}^n$. Let $\alpha_n = \Sigma^{\infty-n} f_n : BO(n) \to \mathcal{S}\mathrm{p}$, then $\alpha_n(*) = \Sigma^{\infty-n} f_n(*) = \Sigma^{\infty-n} S^n \simeq \mathbb{S}$, so $\alpha_n$ factors through $\mathcal{L}\mathrm{ine}_{\mathbb{S}}$. Let $X = BO(n)^{\mathrm{op}}$ and $p : X \to *$ the terminal functor, then

$$M\alpha_n = p_! \Sigma^{\infty-n} f_n \simeq \Sigma^{\infty-n} p_!(f_n)_! y \simeq \Sigma^{\infty-n} (f_n)_! \underbrace{p_!(y)}_{\simeq *} \simeq \Sigma^{\infty-n}(* \times S^n)/O(n)$$

Let $P = EO(n)$ be the universal $O(n)$-bundle and $M$ a $O(n)$-space, then $* \times_{O(n)} M$ is modelled by the *strict* quotient $(P \times M)/O(n)$, then

$$S^n/O(n) = \mathrm{cofib}(\mathbb{R}^n_0 \subseteq \mathbb{R}^n)/O(n) \simeq \mathrm{cofib}(\underbrace{* \times_{O(n)} \mathbb{R}^n_0}_{\simeq E^n_0} \subseteq \underbrace{* \times_{O(n)} \mathbb{R}^n}_{\simeq E^n}) = \mathrm{Th}(E^n)$$

where $E^n = P \times_{O(n)} \mathbb{R}^n \to BO(n)$ is the universal $n$-dimensional vector bundle, hence $M\alpha_n \simeq \Sigma^{\infty-n}\mathrm{Th}(E^n)$.

The functor $BO(n) \to \mathcal{L}\mathrm{ine}_{\mathbb{S}}$ induces a $\infty$-group homomorphism $j_n : O(n) \to GL_1(\mathbb{S})$, mapping $\phi$ to $\Sigma^{\infty-n}\mathrm{Th}(\phi)$. Consider the suspension morphism $s_n = \mathbb{R} \oplus - : O(n) \to O(1+n)$, then

$$j_n(\mathbb{R} \oplus \phi) = \Sigma^{\infty-n-1}\mathrm{Th}(\mathbb{R} \oplus \phi) \simeq \Sigma^{\infty-n-1} \underbrace{\mathrm{Th}(\mathbb{R})}_{\simeq S^1} \wedge \mathrm{Th}(\phi) \simeq \Sigma^{\infty-n}\mathrm{Th}(\phi) = j_n$$

Recall that the colimit over the suspension morphisms $s_n$ is the stable orthogonal group $O$.

**Definition 2.4.** Denote by $j$ the induced group homomorphism $O \to GL_1(\mathbb{S})$, called the *$J$-homomorphism.*

**Example 2.5.** Let $X = O^{\mathrm{op}}$ and take $Bj : BO \to \mathcal{L}\mathrm{ine}_{\mathbb{S}}$, then $Mj$ is denoted $MO$ and called the *real bordism spectrum.*

Denote by $M$ the extended Thom spectrum functor $\mathcal{G}\mathrm{rp}_\infty^{\mathrm{op}}/\mathcal{M}\mathrm{od}_R \to \mathcal{M}\mathrm{od}_R$, this is a left adjoint to the functor sending a $R$-module to the corresponding functor $* \to \mathcal{M}\mathrm{od}_R$. In particular, $M$ preserves colimits and $Bj \simeq \mathrm{colim}_n Bj_n$, therefore we have the following:

**Theorem 2.6.** $MO \simeq \mathrm{colim}_n MO(n) = \mathrm{colim}_n \Sigma^{\infty-n}\mathrm{Th}(E^n)$.

**Example 2.7.** A group homomorphism $\xi : G \to O$ induces a functor $f : BG \to \mathcal{L}\mathrm{ine}_{\mathbb{S}}$. The Thom spectrum $Mf$ is denoted $MG$ or $M\xi$, and called *$G$-bordism spectrum.* For $G = U, SO, Spin$, and $String$, we obtain the *complex, oriented, spin*, and *string bordism spectra.*

*Remark* 2.8. In Equation (2.7) we might take $G = \{*\}$, the one-point group, then $MG \simeq \mathbb{S}$, which, if it didn't have a name, might be called *framed bordism spectrum*, following the naming convention in Equation (2.7) and in line with the theorem that $\pi_*(\mathbb{S}) \simeq \Omega^{\mathrm{fr}}_*$, the bordism ring of framed (trivialized tangent bundle) smooth manifolds.

Let $R$ be a ring in sets, then $R$ is a $A_\infty$-ring spectrum (actually, $E_\infty$), $\Omega^\infty R$ is equivalent to $R$ with discrete topology ($\pi_0(\Omega^\infty R) \simeq R$, as sets, and every other homotopy group vanish). In particular, $GL_1(R)$ is simply $R^\times$ with discrete topology. Consider then the fiber sequence $SO \to O \to \mathbb{Z}^\times \simeq GL_1(\mathbb{Z})$.

**Example 2.9.** $X = BO^{\mathrm{op}}$ and $\alpha = w_1 : BO \to \mathcal{L}\mathrm{ine}_{\mathbb{Z}}$, the 1st Stiefel-Whitney class (delooping of the determinant $O \to GL_1(\mathbb{Z})$), then $Mw_1$ is a $\mathbb{Z}$-module spectra. Let $i : SO \subseteq O$, then $w_1 i$ factors through the point, so $M(w_1 i) \simeq \mathbb{Z} \wedge \Sigma^\infty_+ SO$.

Given $f : X^{\mathrm{op}} \to \mathcal{L}\mathrm{ine}_R$ and a sequence $F \xrightarrow{i} Y \xrightarrow{\pi} X$, there is an induced sequence of Thom $R$-module spectra $MF \to MY \to MX$. If $\pi i$ factors through the point, $MF \simeq R \wedge \Sigma^\infty_+ F$.

**Lemma 2.10.** *Let $R$ be a ring spectrum and $X$ a connected monoidal $\infty$-groupoid, then*

$$\mathrm{Hom}_{\mathcal{M}\mathrm{on}(\mathcal{S}\mathrm{p})}(\Sigma^\infty_+ X, R) \simeq \mathrm{Hom}_{\mathcal{M}\mathrm{on}(\mathcal{S})}(X, GL_1(R))$$

*Proof.* Since $X$ is connected, the space of homomorphisms $X \to GL_1(R)$ is equivalent to the space of homomorphisms $X \to \Omega^\infty R$, then use that $(\Sigma^\infty, \Omega^\infty)$ is a monoidal adjunction (The monoidal structure on spectra is such that $\Sigma^\infty$ is strong monoidal). □

*Remark* 2.11. Notice that we can weaken the result. Namely, if $X$ is 1-connected (pointed and connected), then the space of functors (of $\infty$-groupoids) $X \to GL_1(R)$ is equivalent to the space of functors $X \to \Omega^\infty R$ such that $* \to X \to \Omega^\infty R$ is an equivalence. This last space is equivalent, via the $(\Sigma^\infty_+, \Omega^\infty)$ adjunction, to the space of morphisms of spectra $\Sigma^\infty_+ X \to R$, such that $\mathbb{S} \to \Sigma^\infty_+ X \to R$ represents a unit in $\pi_0(R)$.

*Remark* 2.12. Notice that we can also strengthen the result. Namely, if $X$ is a connected, commutative monoid object and $R$ is a commutative ring spectrum, then $\Omega^\infty R$ and $GL_1(R)$ are also commutative monoid objects. Using the same argument, together with the fact that $(\Sigma^\infty, \Omega^\infty)$ is actually a *symmetric* monoidal adjunction, we conclude that

$$\mathrm{Hom}_{\mathcal{C}\mathcal{M}\mathrm{on}(\mathcal{S}\mathrm{p})}(\Sigma^\infty_+ X, R) \simeq \mathrm{Hom}_{\mathcal{C}\mathcal{M}\mathrm{on}(\mathcal{S})}(X, GL_1(R))$$

*Remark* 2.13. Let $\mathcal{D}$ be a monoidal $\infty$-category. Consider $\mathcal{C}\mathrm{at}_\infty/\mathcal{D}$, the $\infty$-category of functors into $\mathcal{D}$, with monoidal structure given by

$$(F : \mathcal{A} \to \mathcal{D}, G : \mathcal{B} \to \mathcal{D}) \longmapsto (\mathcal{A} \times \mathcal{B} \xrightarrow{F \times G} \mathcal{D} \times \mathcal{D} \xrightarrow{\otimes} \mathcal{D})$$

The monoidal unit is the functor $* \to \mathcal{D}$ picking out the monoidal unit of $\mathcal{D}$. If $\mathcal{D}$ is symmetric monoidal, then so is $\mathcal{C}\mathrm{at}_\infty/\mathcal{D}$. A (commutative) monoid object in $\mathcal{C}\mathrm{at}_\infty/\mathcal{D}$ is given by a (symmetric) monoidal category $\mathcal{C}$ and a (symmetric) monoidal functor $F : \mathcal{C} \to \mathcal{D}$.

In view of Equation (2.13), let $R$ be commutative ring spectrum, then $\mathcal{M}\mathrm{od}_R$ is a symmetric monoidal $\infty$-category and $\mathcal{L}\mathrm{ine}_R$ is a symmetric monoidal $\infty$-groupoid. The category $\mathcal{G}\mathrm{rpd}^{\mathrm{op}}_\infty/\mathcal{L}\mathrm{ine}_R$ is then symmetric monoidal and (commutative) monoid objects are given by (symmetric) monoidal $\infty$-groupoids $X^{\mathrm{op}}$ a (symmetric) monoidal functors $X^{\mathrm{op}} \to \mathcal{L}\mathrm{ine}_R$. One can then check that $M$ is a symmetric monoidal functor, so that (commutative) monoid objects are sent to (commutative) monoid objects in $\mathcal{M}\mathrm{od}_R$, i.e. (commutative) $R$-algebras.

**Example 2.14.** Let Tmf be the commutative ring spectrum of topological modular forms (see Equation (2.15)) and $\sigma : MString \to \mathrm{tmf}$ the *String*-orientation of tmf. In the sequence

$$BString \longrightarrow BO \xrightarrow{Bj} \mathcal{L}\mathrm{ine}_{\mathbb{S}}$$

all functors are symmetric monoidal, so that $MString$ is a commutative $\mathbb{S}$-algebra, i.e. a commutative ring. The *String*-orientation of tmf is also a commutative ring homomorphism. In the fiber sequence $K(\mathbb{Z}, 3) \to BString \to BSO$, the fiber map $i : K(\mathbb{Z}, 3) \to BString$ is also a symmetric monoidal, so the composition

$$\Sigma^\infty_+ K(\mathbb{Z}, 3) \xrightarrow{Mi} MString \xrightarrow{\sigma} \mathrm{tmf}$$

is a commutative ring homomorphism. Using Equation (2.10), we conclude that the induced homomorphism $K(\mathbb{Z}, 3) \to \Omega^\infty \mathrm{tmf}$ (which is a homomorphism of commutative monoid objects, given Equation (2.12)) factors through $GL_1(\mathrm{tmf})$, and so it induces a (symmetric monoidal) functor $K(\mathbb{Z}, 4) \to \mathcal{L}\mathrm{ine}_{\mathrm{tmf}}$, i.e. *2-bundle gerbes twist* tmf.

*Remark* 2.15. The spectrum of topological modular forms comes in three main flavors, namely:

(1) TMF, i.e. the global sections of the spectral structure sheaf $\mathscr{O}^{top} : (\mathrm{Aff}/\mathcal{M}_{ell})^{\mathrm{op}} \to \mathcal{CM}\mathrm{on}(\mathcal{S}\mathrm{p})$ on the (étale site of the) moduli stack of elliptic curves.
(2) Tmf, i.e. the global sections of the spectral structure sheaf $\bar{\mathscr{O}}^{top} : (\mathrm{Aff}/\bar{\mathcal{M}}_{ell})^{\mathrm{op}} \to \mathcal{CM}\mathrm{on}(\mathcal{S}\mathrm{p})$ on the (étale site of the) *compactified* moduli stack of elliptic curves. The inclusion $\mathcal{M}_{ell} \hookrightarrow \bar{\mathcal{M}}_{ell}$ induces a commutative ring homomorphism $\mathrm{Tmf} \to \mathrm{TMF}$.
(3) tmf, i.e. the connective cover of Tmf. By definition, there is a commutative ring homomorphism $\mathrm{tmf} \to \mathrm{Tmf}$.

In [AHR10], tmf is used to denote our TMF, in [Goe09], tmf is used to denoted our Tmf, and [DFHH14] has the same notation as us. In Equation (2.14), we use tmf to mean the connective cover of Tmf.

Let us go down one step in the chromatic ladder.

**Example 2.16.** Recall the fiber sequences for $Spin$ and $Spin^c$:

$$\mathbb{Z}_2 \to Spin \to SO, \qquad S^1 \to Spin^c \to SO$$

All the spaces involved are commutative groups. Applying the Thom spectrum functor to the delooped sequences, we get

$$\Sigma^\infty_+ K(\mathbb{Z}_2, 1) \to MSpin \to MSO, \qquad \Sigma^\infty_+ K(\mathbb{Z}, 2) \to MSpin^c \to MSO$$

Let $\sigma : MSpin \to KO$ and $\sigma^c : MSpin^c \to KU$ be the Atiyah-Bott-Shapiro orientation of real and complex $K$-theory (see [ABS64]). Similar to Equation (2.14), we get homomorphisms

$$\Sigma^\infty_+ K(\mathbb{Z}_2, 1) \longrightarrow MSpin \xrightarrow{\sigma} KO, \qquad \Sigma^\infty_+ K(\mathbb{Z}, 2) \longrightarrow MSpin^c \xrightarrow{\sigma^c} KU$$

Using Equation (2.10) again and delooping, we obtain functors $K(\mathbb{Z}_2, 2) \to \mathcal{L}\mathrm{ine}_{KO}$ and $K(\mathbb{Z}, 3) \to \mathcal{L}\mathrm{ine}_{KU}$, i.e. *real, resp. complex, bundle gerbes twist real, resp. complex, $K$-theory.*

## 3. Twists via Picard Groupoids and Grading

This section requires some some further details. Up until now we defined everything via $\mathcal{L}\mathrm{ine}_R$, however for many applications we need to work with $\mathcal{P}\mathrm{ic}_R$ instead.

**Definition 3.1.** Given a monoidal $\infty$-category $(\mathcal{C}, \otimes, 1)$, an object $M$ is *invertible* if there is an object $N$ such that $N \otimes M \simeq M \otimes N \simeq 1$. The *Picard $\infty$-groupoid* of $\mathcal{C}$ is the sub-$\infty$-groupoid generated by invertible modules.

*Remark* 3.2. A monoidal category $(\mathcal{C}, \otimes, 1)$ is *closed* if, for every object $M$, the *left tensoring with $M$* functor $M^\otimes : \mathcal{C} \to \mathcal{C}$ admits a right adjoint $F(M, -)$. If $M$ is invertible, with inverse $N$, then $M^\otimes$ is an equivalence with $N^\otimes$ as inverse. In particular, we can promote $(M^\otimes, N^\otimes)$ to an adjoint equivalence. By uniqueness of adjoint functors $F(M, -) \simeq N^\otimes$, and so

$$N \simeq N \otimes 1 = N^\otimes(1) \simeq F(M, 1) =: DM$$

**Definition 3.3.** Given a closed monoidal category $(\mathcal{C}, \otimes, 1)$, the functor $D := F(-, 1) : \mathcal{C}^{\mathrm{op}} \to \mathcal{C}$ will be called *duality*. Given an object $M$, the object $DM$ is called the *dual of $M$*.

**Definition 3.4.** If $R$ is a ring spectrum, $\mathcal{M}\mathrm{od}_R$ is closed monoidal. Denote by $\mathcal{P}\mathrm{ic}_R$ the Picard $\infty$-groupoid of $\mathcal{M}\mathrm{od}_R$.

*Remark* 3.5. $\Sigma^n R$ is invertible, with inverse $\Sigma^{-n} R$. In particular, there is a map $\mathbb{Z} \times \mathcal{L}\mathrm{ine}_R \to \mathcal{P}\mathrm{ic}_R$. However, this map need not be neither injective (if $R$ is $n$-periodic), nor surjective (see [HM17]).

As mentioned in, the Thom spectrum functor makes sense for every functor $f : X^{\mathrm{op}} \to \mathcal{M}\mathrm{od}_R$. However, all examples of twists encountered so far came from functors into $\mathcal{L}\mathrm{ine}_R$. An example of twist that is not the result of a $R$-line bundle is the *degree shift*.

**Definition 3.6.** Denote by $M$ the *Thom $R$-module spectrum* functor

$$\mathcal{G}\mathrm{rpd}^{\mathrm{op}}_\infty / \mathcal{M}\mathrm{od}_R \to \mathcal{M}\mathrm{od}_R$$

sending a functor $f : X^{\mathrm{op}} \to \mathcal{M}\mathrm{od}_R$ to its colimit.

**Example 3.7.** Let $f : X^{\mathrm{op}} \to \mathcal{L}\mathrm{ine}_R$ be a twist. Denote by $\Sigma^n f$ the composition of $f$ with the shift functor $\Sigma^n : \mathcal{L}\mathrm{ine}_R \to \mathcal{P}\mathrm{ic}_R$. Since $\Sigma^n$ is an equivalence, it commutes with colimits, so

$$M\Sigma^n f \simeq \Sigma^n M f$$

If $f = R_X$, then $M\Sigma^n f \simeq \Sigma^n R \wedge \Sigma^\infty_+ X$, so $\Sigma^n f$-twisted $R$-cohomology and $R$-homology correspond to normal $R$-cohomology and $R$-homology with a degree shift by $n$.

## 4. Umkehr Map

We now proceed to the construction of the umkehr map in twisted cohomology theories. Here we follow [ABG18]. Let $R$ be a ring spectrum, denote by $D_R$ the duality of $\mathcal{M}\mathrm{od}_R$. Given a invertible $R$-twist $\alpha : X^{\mathrm{op}} \to \mathcal{P}\mathrm{ic}_R$, denote by $D_R\alpha$ the post-composition of $\alpha^{\mathrm{op}}$ with $D_R$.

*Remark* 4.1. Depending on which cohomological degrees we want, we sometimes use the following alternative definition of twisted cohomology:

$$R^\alpha(X) = \pi_0 \mathrm{Hom}_{\mathcal{M}\mathrm{od}_R}(M(D_R\alpha), R)$$

meaning we use the dual/inverse of $\alpha$. To justify the use of $D_R\alpha$, consider the following: Let $\beta = D_R\alpha$, then

$$\mathrm{Hom}_{\mathcal{M}\mathrm{od}_R}(M\beta, R) \simeq \mathrm{Hom}_{\mathcal{M}\mathrm{od}_R(X)}(D_R\alpha, R_X) \simeq \mathrm{Hom}_{\mathcal{M}\mathrm{od}_R(X)}(R_X, \alpha \otimes_{R_X} R_X) \simeq \mathrm{Hom}_{\mathcal{M}\mathrm{od}_R(X)}(R_X, \alpha)$$

The second equivalence is a consequence of the fact that, if $M$ be an invertible $R$-module, left tensoring with $M$ is left adjoint to tensoring with $D_R M$, and $D_R D_R M \simeq M$, see Equation (3.2). If we think of $\alpha$ as a bundle of invertible $R$-modules over $X$, then $\mathrm{Hom}_{\mathcal{M}\mathrm{od}_R(X)}(R_X, \alpha)$ is the spectrum of global sections of $\alpha$, which aligns with the idea that twisted cohomology is the homotopy groups of the global sections of a bundle of spectra.

If $R = \mathbb{S}$, we denote the duality $D_R$ by simply $D$.

**Definition 4.2.** The *Spanier-Whitehead dual* of a space $X$ is the dual spectrum of $\Sigma^\infty_+ X$ with respect to the sphere spectrum.

*Remark* 4.3. Recall that $\mathcal{S}\mathrm{p}$ is a closed symmetric monoidal category, with $\mathcal{H}\mathrm{om}_{\mathcal{S}\mathrm{p}}(E, -)$ being the right adjoint to the functor given by smash product with $E$. Let $(\mathcal{C}, \otimes, 1)$ a closed monoidal category, where $F(X, -)$ is the right adjoint to the functor given by left tensoring with $X$, then

$$hom_{\mathcal{C}}(-, F(1, E)) \simeq \mathrm{Hom}_{\mathcal{C}}(1 \otimes -, E) \simeq \mathrm{Hom}_{\mathcal{C}}(-, E)$$

Using Yoneda's lemma, we conclude that $E \simeq F(1, E)$, for all $E$.

**Example 4.4.** By Equation (4.3), the Spanier-Whitehead dual to $*$ is the sphere spectrum, since

$$D* = \mathcal{H}\mathrm{om}(\Sigma^\infty_+ *, \mathbb{S}) = \mathcal{H}\mathrm{om}(\Sigma^\infty S^0, \mathbb{S}) = \mathcal{H}\mathrm{om}(\mathbb{S}, \mathbb{S}) \simeq \mathbb{S}$$

**Definition 4.5.** Denote by $\phi$ the functor $\mathcal{S}^{\mathrm{op}} \to \mathbb{S}/\mathcal{S}\mathrm{p}$ from spaces to the category of spectra under $\mathbb{S}$, mapping $X$ to the map of spectra

$$\phi(X) : \mathbb{S} \simeq D* \xrightarrow{Dp} DX$$

where $p$ is the terminal map $X \to *$.

**Definition 4.6.** Denote by $\widehat{\mathcal{M}\mathrm{fd}}_\infty$ the topological groupoid of closed smooth manifolds and diffeomorphisms. The set of diffeomorphisms are topologized by the weak $C^\infty$ topology, see [Hir94]. Let $\mathcal{M}\mathrm{fd}_\infty$ be homotopy coherent nerve of $\widehat{\mathcal{M}\mathrm{fd}}_\infty$.

Let $B$ be a connected compact space and $\pi\colon X \to B$ a continuous function such that, for every $b \in B$, the fiber $\pi^{-1}(b) =: X_b$ is equipped with the structure of a closed smooth manifold, which varies continuously in $B$, in the sense of being classified by a functor

$$f : B \to \mathcal{M}\mathrm{fd}_\infty$$

Given such a classifying map $f$, consider the following composition

$$B^{\mathrm{op}} \longrightarrow \mathcal{M}\mathrm{fd}_\infty^{\mathrm{op}} \longrightarrow \mathcal{S}^{\mathrm{op}} \xrightarrow{\phi} \mathbb{S}/\mathcal{S}\mathrm{p}$$

where the middle functor is the one forgetting the smooth structure. The above composition is then an object of

$$\mathcal{F}\mathrm{un}(B^{\mathrm{op}}, \mathbb{S}/\mathcal{S}\mathrm{p}) \simeq \mathbb{S}_B/\mathcal{F}\mathrm{un}(B^{\mathrm{op}}, \mathcal{S}\mathrm{p})$$

i.e. it is a natural transformation

$$\phi_{X/B} : \mathbb{S}_B \to D_B(f)$$

where $D_B$ denotes the Spanier-Whithead duality applied to (the opposite of) $f : B \to \mathcal{S}$ point-wise. Let $p : B \to *$ be the terminal functor and $p_! : \mathcal{F}\mathrm{un}(B^{\mathrm{op}}, \mathcal{S}\mathrm{p}) \to \mathcal{S}\mathrm{p}$ the left Kan extension along $p$, i.e. the colimit functor. Applying $p_!$ to $\phi_{X/B}$, we obtain a morphism of spectra

$$\Sigma^\infty_+ B \simeq p_! p^* \mathbb{S} = p_! \mathbb{S}_B \xrightarrow{p_!(\phi_{X/B})} p_! D_B(f) \tag{4.7}$$

Let $T_{X/B} \to X$ be the vector bundle of fiber-wise tangent vectors, classified by $X^{\mathrm{op}} \to BO(n)$, where $n$ is the manifold dimension of the fibers[1]. Denote by $\alpha$ the composition

$$X^{\mathrm{op}} \xrightarrow{T_{X/B}} BO(n) \longrightarrow \mathcal{S}_* \xrightarrow{\Sigma^\infty} \mathcal{S}\mathrm{p} \tag{4.8}$$

the middle arrow being the $n$-sphere $S^n$ with $O(n)$-action coming from the one-point compactification of the regular action on $\mathbb{R}^n$. The above diagram takes values in the Picard $\infty$-groupoid of the sphere spectrum.

**Theorem 4.9.** $p_! D_B(f) \simeq M(-\alpha) =: X^{-T_{X/B}}$.

**Definition 4.10.** The morphism

$$\mathcal{PT}(f) : \Sigma^\infty_+ B \longrightarrow X^{-T_{X/B}}$$

is called *Pontryagin-Thom transfer map.*

Consider now a general situation. Let $R$ be a ring spectrum and $\alpha : X^{\mathrm{op}} \to \mathcal{L}\mathrm{ine}_R$ a $R$-line bundle.

**Definition 4.11.** An *orientation* of $M\alpha$ is lift

$$\begin{tikzcd} & * \arrow[d] \\ X^{\mathrm{op}} \arrow[ur, dashed] \arrow[r, "\alpha"'] & \mathcal{L}\mathrm{ine}_R \end{tikzcd}$$

Explicitly, it is an equivalence of $R_X$-modules $\alpha \to R_X$.

Applying $p_!$, we see that an equivalence $t : \alpha \to R_X$ induces an equivalence

$$M\alpha \xrightarrow{p_!(t)} R \wedge \Sigma^\infty_+ X$$

In our case, $\alpha$ in Equation (4.8) is not valued in $\mathcal{L}\mathrm{ine}_{\mathbb{S}}$, but $\Sigma^{-n}\alpha$ is. An orientation of $\Sigma^{-n}\alpha$ induces an equivalence

$$\Sigma^{-n} M\alpha \xrightarrow{\simeq} R \wedge \Sigma^\infty_+ X \tag{4.12}$$

**Definition 4.13.** The isomorphism in cohomology induced by Equation (4.12) is called *Thom isomorphism.*

Finally, we have the following definition:

**Definition 4.14.** Assuming $\Sigma^{-n}\alpha : X^{\mathrm{op}} \to \mathcal{L}\mathrm{ine}_R$ is orientable, the *Umkehr map* is the map:

$$R^*(\Sigma^\infty_+ X) \xrightarrow{\text{Thom iso.}} R^{*-n}(X^{-T_{X/B}}) \xrightarrow{\mathcal{PT}(f)^{*-n}} R^{*-n}(\Sigma^\infty_+ B)$$

*Remark* 4.15. The name *Umkehr* comes from the map going in the opposite direction to $R^*(\Sigma^\infty_+ X) \to R^*(\Sigma^\infty_+ B)$, given by pulling back along $\pi$.

We now introduce the *twisted Umkehr map.* Given a twist $\beta\colon B^{\mathrm{op}} \to \mathcal{P}\mathrm{ic}_R$, we can smash to get a map

$$\beta \simeq \mathbb{S}_B \wedge_B \beta \longrightarrow D_B(f) \wedge_B \beta$$

Applying the functor $p_!$ once again, we obtain a morphism

$$M\beta \longrightarrow p_!(D_B(f) \wedge_B \beta) \simeq X^{-T_f + \beta\pi}$$

[1] $B$ is connected, so the fibers have constant dimension.

where $\beta\pi : X^{\mathrm{op}} \to \mathcal{P}\mathrm{ic}_R$ is the composition of $\pi$ and the twist $\beta$.

**Definition 4.16.** The *twisted Pontryagin-Thom transfer map* is the morphism:

$$\mathcal{PT}(f,\beta) : M\beta \longrightarrow X^{-T_f+\beta\pi}$$

The *twisted Umkehr map* is the map of twisted cohomology groups induced by $\mathcal{PT}(f,\beta)$.

**Example 4.17.** Assume that $X^{-T_{X/B}+\beta\pi}$ is orientable, i.e. the following diagram commutes

$$\begin{tikzcd} X \arrow[r, "-T_f"] \arrow[d] & \mathcal{P}\mathrm{ic}_{\mathbb{S}} \arrow[d] \\ B \arrow[r, "\alpha"] & \mathcal{P}\mathrm{ic}_R \end{tikzcd}$$

then $X^{-T_{X/B}+\beta\pi} \simeq R \wedge \Sigma^\infty_+ X$ and the twisted Umkehr map becomes:

$$R^*(\Sigma^\infty_+ X) \longrightarrow R^{*-\beta}(B)$$

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